SZABIST
University

Shaheed Zulfikar Ali Bhutto
Institute of Science and Technology

ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT

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Total Marks:________

Obtained Marks: ________

Computing Vision

Lab Task # 14

Submitted To: **Mr. Muhammad Azhar**

Student Name: **M. Mateen Bashir**

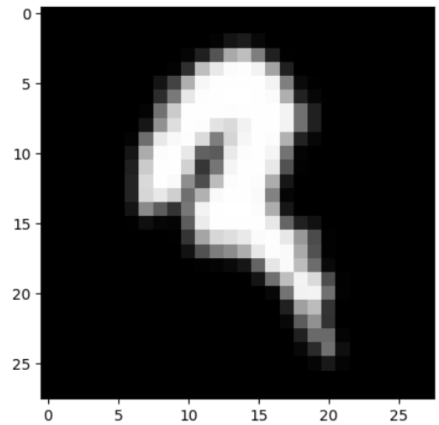
Reg. Number: **23108125**

Code Example 1: Load and Preprocess Character Dataset
Code Example 2: Neural Network for Character Recognition
Code Example 3: Model Evaluation and Prediction

```
# Code Example 1: Load and Preprocess Character Dataset
# Code Example 2: Neural Network for Character Recognition
# Code Example 3: Model Evaluation and Prediction
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense
import struct

def load_images(file):
    with open(file, 'rb') as f:
        f.read(16)
        data = np.frombuffer(f.read(), dtype=np.uint8)
    return data.reshape(-1, 28, 28, 1)

def load_labels(file):
    with open(file, 'rb') as f:
        f.read(8)
        labels = np.frombuffer(f.read(), dtype=np.uint8)
    return labels

X_train = load_images("D:/EMNIST/emnist_source_files/emnist-letters-train-images-idx3-ubyte")
y_train = load_labels("D:/EMNIST/emnist_source_files/emnist-letters-train-labels-idx1-ubyte")

X_test = load_images("D:/EMNIST/emnist_source_files/emnist-letters-test-images-idx3-ubyte")
y_test = load_labels("D:/EMNIST/emnist_source_files/emnist-letters-test-labels-idx1-ubyte")

X_train = X_train / 255.0
X_test = X_test / 255.0

y_train = to_categorical(y_train - 1, 26)
y_test = to_categorical(y_test - 1, 26)

model = Sequential([
    Conv2D(32, (3,3), activation='relu', input_shape=(28,28,1)),
    MaxPooling2D(2,2),
    Conv2D(64, (3,3), activation='relu'),
    MaxPooling2D(2,2),
    Flatten(),
```

```
    Dense(128, activation='relu'),
    Dense(26, activation='softmax')
])

model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

model.fit(X_train, y_train, epochs=3, validation_split=0.2)

loss, acc = model.evaluate(X_test, y_test)
print("Accuracy:", acc * 100)

labels = list("ABCDEFGHIJKLMNOPQRSTUVWXYZ")

plt.imshow(X_test[0].reshape(28,28), cmap='gray')
plt.show()

pred = model.predict(X_test[0].reshape(1,28,28,1))
print("Predicted:", labels[np.argmax(pred)])
```

```
Epoch 1/3
3120/3120 ━━━━━━━━━━━━━━━━━━━━ 28s 9ms/step - accuracy: 0.8549 - loss: 0.4646 - val_accuracy: 0.9034 - val_loss: 0.2969
Epoch 2/3
3120/3120 ━━━━━━━━━━━━━━━━━━━━ 26s 8ms/step - accuracy: 0.9209 - loss: 0.2391 - val_accuracy: 0.9184 - val_loss: 0.2526
Epoch 3/3
3120/3120 ━━━━━━━━━━━━━━━━━━━━ 26s 8ms/step - accuracy: 0.9353 - loss: 0.1900 - val_accuracy: 0.9308 - val_loss: 0.2116
650/650 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.9293 - loss: 0.2154
Accuracy: 92.92788505554199
```




```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 90ms/step
Predicted: A
```

Code Example 4: Handwritten Alphabet Recognition

An educational technology company is developing an app to assist young children and individuals with writing difficulties in learning and practicing handwriting. A key feature of the app is to automatically recognize handwritten alphabetical characters (A-Z) entered via a touchscreen or stylus.

The challenge is to accurately recognize diverse handwriting styles, varying stroke thicknesses, and distortions common in natural handwriting. Traditional rule-based systems proved ineffective due to the wide variability in handwriting.

```python
# Code Example 4: Handwritten Alphabet Recognition
import tensorflow as tf
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import cv2

from sklearn.model_selection import train_test_split
from keras.models import Sequential
from keras.layers import Conv2D
from keras.layers import MaxPooling2D
from keras.layers import Dense
from keras.layers import Dropout
from keras.layers import Flatten
from keras.layers import BatchNormalization
from keras.optimizers import SGD

train_df = pd.read_csv('D:/EMNIST/emnist-letters-train.csv')
test_df = pd.read_csv('D:/EMNIST/emnist-letters-test.csv')

df = pd.concat([train_df, test_df], ignore_index=True)

data_array = np.array(df, dtype=np.uint8)

del df

print(data_array.shape)

alpha = list('ABCDEFGHIJKLMNOPQRSTUVWXYZ')

labels = data_array[:, 0]

valid_idx = labels > 0

labels = labels[valid_idx] - 1

x = data_array[:, 1:785]

x = x[valid_idx]
```

```
x = x.reshape(-1, 28, 28)

x = np.array([np.fliplr(np.rot90(img)) for img in x])

x = x / 255.0

del data_array

unique, counts = np.unique(labels, return_counts=True)

print(unique)
print(counts)

fig = plt.figure(figsize=(15,6))

plt.xlabel('ALPHABETS', fontsize=14)
plt.ylabel('COUNT', fontsize=14)

plt.bar(alpha, counts)

plt.show()

a = np.random.randint(
    low=0,
    high=x.shape[0],
    size=400
)

fig = plt.figure(figsize=(30,30))

c = 1

for i in a:

    fig.add_subplot(20,20,c)

    plt.xticks([])
    plt.yticks([])

    plt.imshow(x[i], cmap='gray')
```

```
    c += 1

plt.show()

x = x.reshape(x.shape[0], 28, 28, 1)

x_train, x_test, y_train, y_test = train_test_split(
    x,
    labels,
    test_size=0.01,
    random_state=42
)

print(x_train.shape)
print(x_test.shape)
print(y_train.shape)
print(y_test.shape)

model = Sequential([

    Conv2D(
        128,
        (3,3),
        activation='relu',
        kernel_initializer='he_uniform',
        input_shape=(28,28,1),
        padding='same'
    ),

    Conv2D(
        64,
        (3,3),
        activation='relu',
        kernel_initializer='he_uniform',
        padding='same'
    ),

    MaxPooling2D(2,2),

    Conv2D(
```

```
        64,
        (3,3),
        activation='relu',
        kernel_initializer='he_uniform',
        padding='same'
    ),

    Conv2D(
        64,
        (3,3),
        activation='relu',
        kernel_initializer='he_uniform',
        padding='same'
    ),

    BatchNormalization(),

    MaxPooling2D(2,2),

    Flatten(),

    Dense(
        100,
        activation='relu',
        kernel_initializer='he_uniform'
    ),

    Dropout(0.1),

    Dense(
        64,
        activation='relu',
        kernel_initializer='he_uniform'
    ),

    Dropout(0.125),

    BatchNormalization(),

    Dense(
```

```
        26,
        activation='softmax'
    )
])

optimizer = SGD(
    learning_rate=0.01,
    momentum=0.9
)

model.compile(
    loss='sparse_categorical_crossentropy',
    optimizer=optimizer,
    metrics=['accuracy']
)

model.summary()

history = model.fit(
    x_train,
    y_train,
    epochs=3,
    batch_size=150,
    validation_data=(x_test, y_test)
)

val_loss = history.history['val_loss']
val_accuracy = history.history['val_accuracy']

loss = history.history['loss']
accuracy = history.history['accuracy']

fig = plt.figure(figsize=(10,15))

fig.add_subplot(2,1,1)

plt.title('Cross Entropy Loss')

plt.plot(loss, color='blue', label='train')
plt.plot(val_loss, color='red', label='test')
```

```
plt.legend()

fig.add_subplot(2,1,2)

plt.title('Classification Accuracy')

plt.plot(accuracy, color='blue', label='train')
plt.plot(val_accuracy, color='red', label='test')

plt.legend()

plt.show()

metrics = model.evaluate(x_test, y_test)

print("Test Accuracy is : {:.2f}".format(metrics[1] * 100))
print("Test Loss is : {:.2f}".format(metrics[0]))

model.save('Alphabet_Recognition.h5')

model = tf.keras.models.load_model('Alphabet_Recognition.h5')

metrics = model.evaluate(x_test, y_test)

print(metrics)

print("Test Accuracy is : {:.2f}".format(metrics[1] * 100))
print("Test Loss is : {:.2f}".format(metrics[0]))

def test_images(n=225):

    index = np.random.randint(
        low=0,
        high=x_test.shape[0],
        size=n
    )

    fig = plt.figure(figsize=(30,40))

    for i in range(n):
```

```
        prediction = model.predict(
            x_test[index[i]].reshape(1,28,28,1),
            verbose=0
        )

        pred = np.argmax(prediction)

        actual = y_test[index[i]]

        fig.add_subplot(15,15,i+1)

        plt.xticks([])
        plt.yticks([])

        if actual == pred:

            plt.title(
                alpha[pred],
                color='green',
                fontsize=25,
                fontweight='bold'
            )

        else:

            plt.title(
                alpha[pred],
                color='red',
                fontsize=25,
                fontweight='bold'
            )

        plt.imshow(
            x_test[index[i]].reshape(28,28),
            cmap='gray'
        )

    plt.show()

test_images()
```

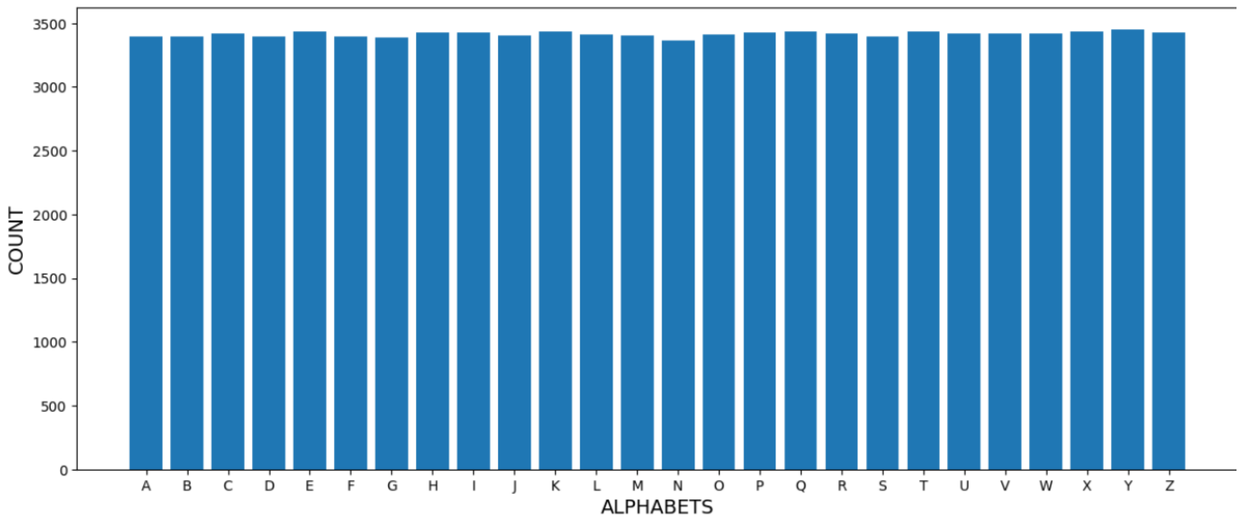
```python
def alphabet_recognize(filepath):

    image = cv2.imread(filepath)

    if image is None:
        print("Image not found")
        return

    blur_image = cv2.medianBlur(image, 7)

    grey = cv2.cvtColor(
        blur_image,
        cv2.COLOR_BGR2GRAY
    )

    thresh = cv2.adaptiveThreshold(
        grey,
        255,
        cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV,
        41,
        15
    )

    contours, hierarchy = cv2.findContours(
        thresh,
        cv2.RETR_EXTERNAL,
        cv2.CHAIN_APPROX_SIMPLE
    )

    if len(contours) == 0:
        print("No alphabets detected")
        return

    preprocessed_digits = []

    boundingBoxes = [cv2.boundingRect(c) for c in contours]

    contours, boundingBoxes = zip(
        *sorted(
```

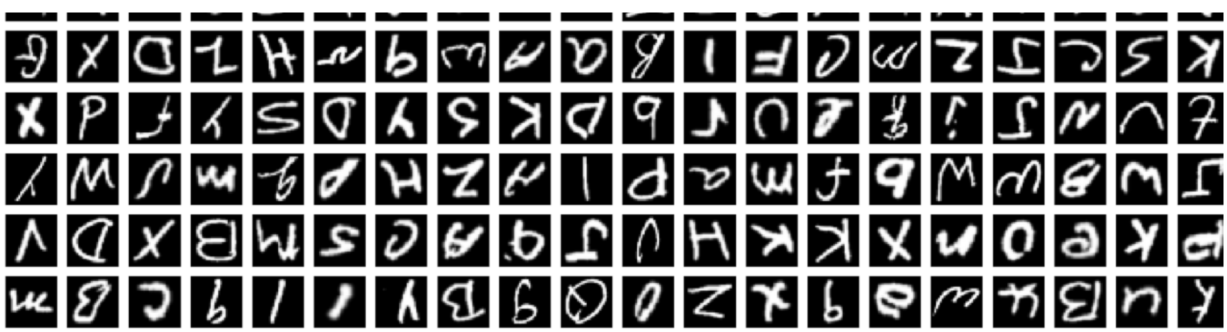
```python
            zip(contours, boundingBoxes),
            key=lambda b: b[1][0],
            reverse=False
        )
    )

    for c in contours:

        x1, y1, w, h = cv2.boundingRect(c)

        cv2.rectangle(
            image,
            (x1, y1),
            (x1 + w, y1 + h),
            (255, 0, 0),
            2
        )

        digit = thresh[y1:y1+h, x1:x1+w]

        resized_digit = cv2.resize(
            digit,
            (18,18)
        )

        padded_digit = np.pad(
            resized_digit,
            ((5,5),(5,5)),
            "constant",
            constant_values=0
        )

        preprocessed_digits.append(padded_digit)

    plt.figure(figsize=(10,10))

    plt.xticks([])
    plt.yticks([])

    plt.title("Contoured Image")
```

```python
    plt.imshow(image)

    plt.show()

    fig = plt.figure(figsize=(20,4))

    alphabets = []

    for i, digit in enumerate(preprocessed_digits):

        prediction = model.predict(
            digit.reshape(1,28,28,1) / 255.0,
            verbose=0
        )

        pred = alpha[np.argmax(prediction)]

        alphabets.append(pred)

        fig.add_subplot(1, len(preprocessed_digits), i+1)

        plt.xticks([])
        plt.yticks([])

        plt.imshow(
            digit.reshape(28,28),
            cmap='gray'
        )

        plt.title(
            pred,
            color='green',
            fontsize=18,
            fontweight='bold'
        )

    plt.show()

    print("The Recognized Alphabets are :", *alphabets)
```

```python
alphabet_recognize('D:/EMNIST/1.jpg')

alphabet_recognize('D:/EMNIST/2.jpg')

alphabet_recognize('D:/EMNIST/3.jpg')
```

```
(103598, 1003)
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
 24 25]
[3396 3396 3419 3398 3437 3394 3385 3424 3428 3402 3438 3415 3402 3365
 3408 3430 3435 3419 3392 3436 3419 3422 3422 3437 3453 3427]
```





```
(87911, 28, 28, 1)
(888, 28, 28, 1)
(87911,)
(888,)
```

Model: "sequential_1"

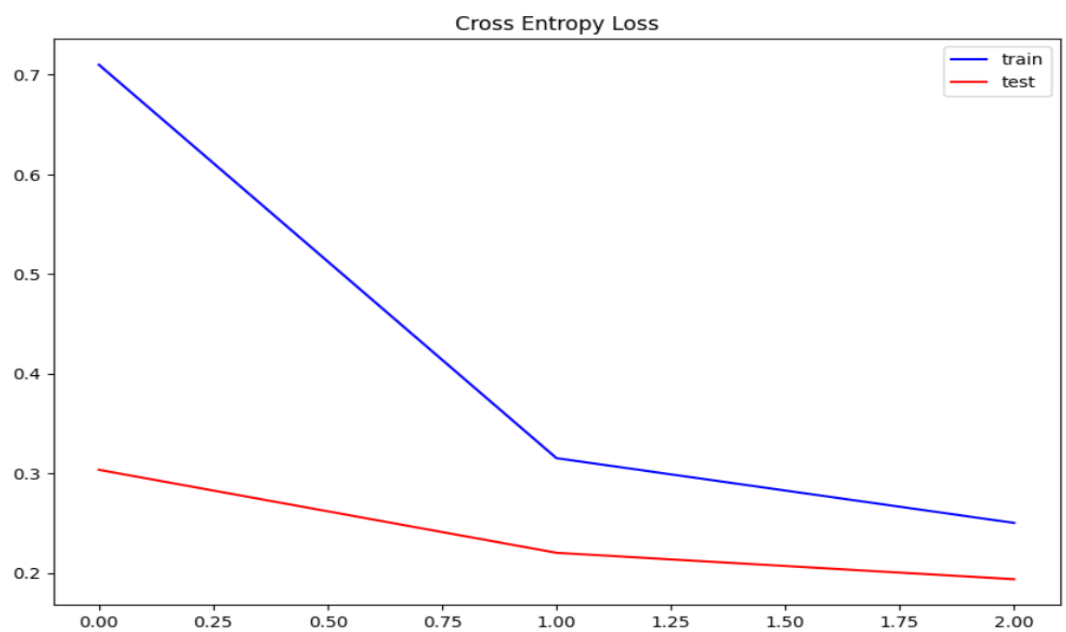
Layer (type)	Output Shape	Param #
conv2d_2 (Conv2D)	(None, 28, 28, 128)	1,280
conv2d_3 (Conv2D)	(None, 28, 28, 64)	73,792
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 64)	0
conv2d_4 (Conv2D)	(None, 14, 14, 64)	36,928
conv2d_5 (Conv2D)	(None, 14, 14, 64)	36,928
batch_normalization (BatchNormalization)	(None, 14, 14, 64)	256
max_pooling2d_3 (MaxPooling2D)	(None, 7, 7, 64)	0
flatten_1 (Flatten)	(None, 3136)	0
dense_2 (Dense)	(None, 100)	313,700
dropout (Dropout)	(None, 100)	0
dense_3 (Dense)	(None, 64)	6,464
dropout_1 (Dropout)	(None, 64)	0
batch_normalization_1 (BatchNormalization)	(None, 64)	256
dense_4 (Dense)	(None, 26)	1,690

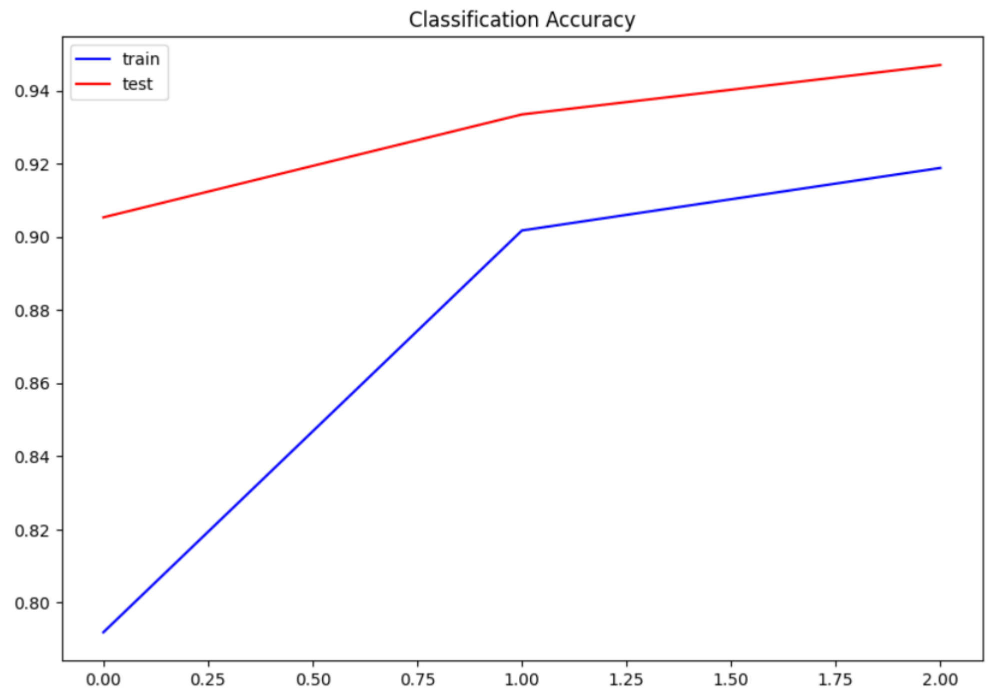
Total params: 471,294 (1.80 MB)

Trainable params: 471,038 (1.80 MB)

Non-trainable params: 256 (1.00 KB)

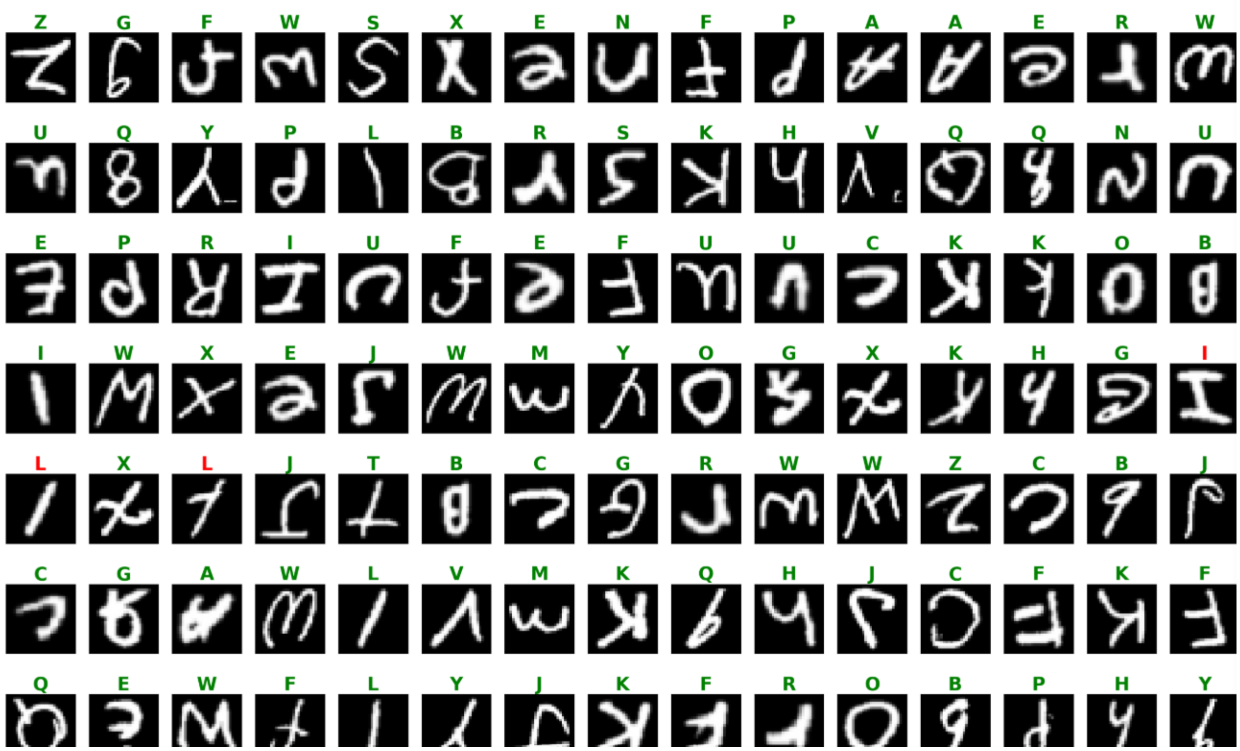
```
Epoch 1/3
587/587 ━━━━━━━━━━━━━━━━━━━━ 238s 403ms/step - accuracy: 0.7919 - loss: 0.7101 - val_accuracy: 0.9054 - val_loss: 0.3037
Epoch 2/3
587/587 ━━━━━━━━━━━━━━━━━━━━ 246s 419ms/step - accuracy: 0.9018 - loss: 0.3154 - val_accuracy: 0.9336 - val_loss: 0.2205
Epoch 3/3
587/587 ━━━━━━━━━━━━━━━━━━━━ 254s 433ms/step - accuracy: 0.9189 - loss: 0.2505 - val_accuracy: 0.9471 - val_loss: 0.1940
```



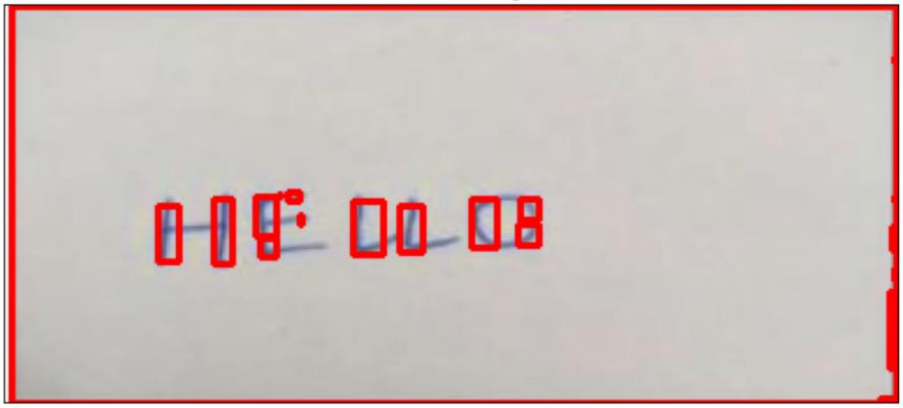



```
28/28 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step - accuracy: 0.9471 - loss: 0.1940
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered leg
We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`.
Test Accuracy is : 94.71
Test Loss is : 0.19
```

```
28/28 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step - accuracy: 0.9471 - loss: 0.1940
[0.19399066269397736, 0.9470720887184143]
Test Accuracy is : 94.71
Test Loss is : 0.19
```



Contoured Image



The Recognized Alphabets are : I O Z S Q N A N T T J L I R E M N N N

Contoured Image

The Recognized Alphabets are : R L O O T R N W O N A N A S L D N N R

Contoured Image

The Recognized Alphabets are : N Q N A N A A V N L O D U A A O Q O N A A Q R

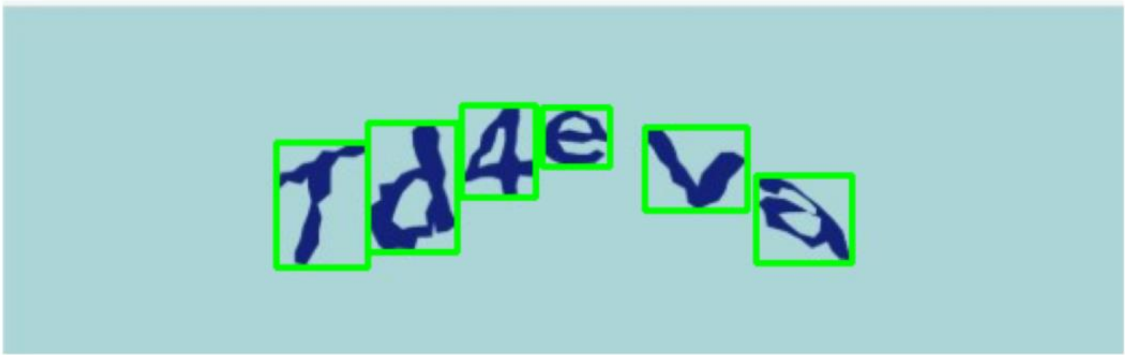
Case Study 1: CAPTCHA Recognition for Automated Bot Detection

Scenario: A cybersecurity company aims to improve its automated bot detection system by enhancing the ability to solve visual CAPTCHAs, which consist of distorted alphanumeric characters designed to differentiate humans from bots. The challenge is that the distortions and noise in CAPTCHAs vary widely, making rule-based or simple OCR techniques unreliable.

Objective: Develop a character recognition model using neural networks that can robustly recognize distorted alphabetical characters in CAPTCHAs to automate validation processes while improving bot detection accuracy.

```python
# Case Study 1: CAPTCHA Recognition for Automated Bot Detection
import cv2
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

model = tf.keras.models.load_model('Alphabet_Recognition.h5')

alpha = list('ABCDEFGHIJKLMNOPQRSTUVWXYZ')

def preprocess_char(img):
    img = cv2.resize(img, (28, 28))
    img = img.astype("float32") / 255.0
    img = img.reshape(1, 28, 28, 1)
    return img

def captcha_recognize(filepath):
    image = cv2.imread(filepath)

    if image is None:
        print("Image not found!")
        return

    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    blur = cv2.GaussianBlur(gray, (5, 5), 0)

    thresh = cv2.adaptiveThreshold(
        blur, 255,
        cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV,
        11, 2
    )

    contours, _ = cv2.findContours(
        thresh, cv2.RETR_EXTERNAL,
        cv2.CHAIN_APPROX_SIMPLE
    )

    contours = sorted(contours, key=lambda c: cv2.boundingRect(c)[0])

    result = []
```

```python
    result = []

    for c in contours:
        x, y, w, h = cv2.boundingRect(c)

        if w < 8 or h < 8:
            continue

        char_img = thresh[y:y+h, x:x+w]

        processed = preprocess_char(char_img)

        prediction = model.predict(processed, verbose=0)
        pred = alpha[np.argmax(prediction)]

        result.append(pred)

        cv2.rectangle(image, (x, y), (x+w, y+h), (0, 255, 0), 2)

    result_text = ''.join(result)

    plt.figure(figsize=(12, 4))
    plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
    plt.axis('off')
    plt.show()

    print("Predicted CAPTCHA:", result_text)


captcha_recognize("D:/EMNIST/Captcha.jpg")
```



Predicted CAPTCHA: QPYANQ

Case Study 2: License Plate Readers in Traffic Management
Scenario: A smart city initiative implements automated license plate readers (LPRs) across highways and urban roads to manage toll collection, enforce traffic laws, and enhance security monitoring. Accurately recognizing alphanumeric characters on license plates, often under challenging lighting, angles, and dirt conditions, is critical for system reliability.
Objective: Design and deploy a neural network-based alphabet recognition system that can correctly identify characters from images of license plates in real time, supporting tolling and law enforcement operations.

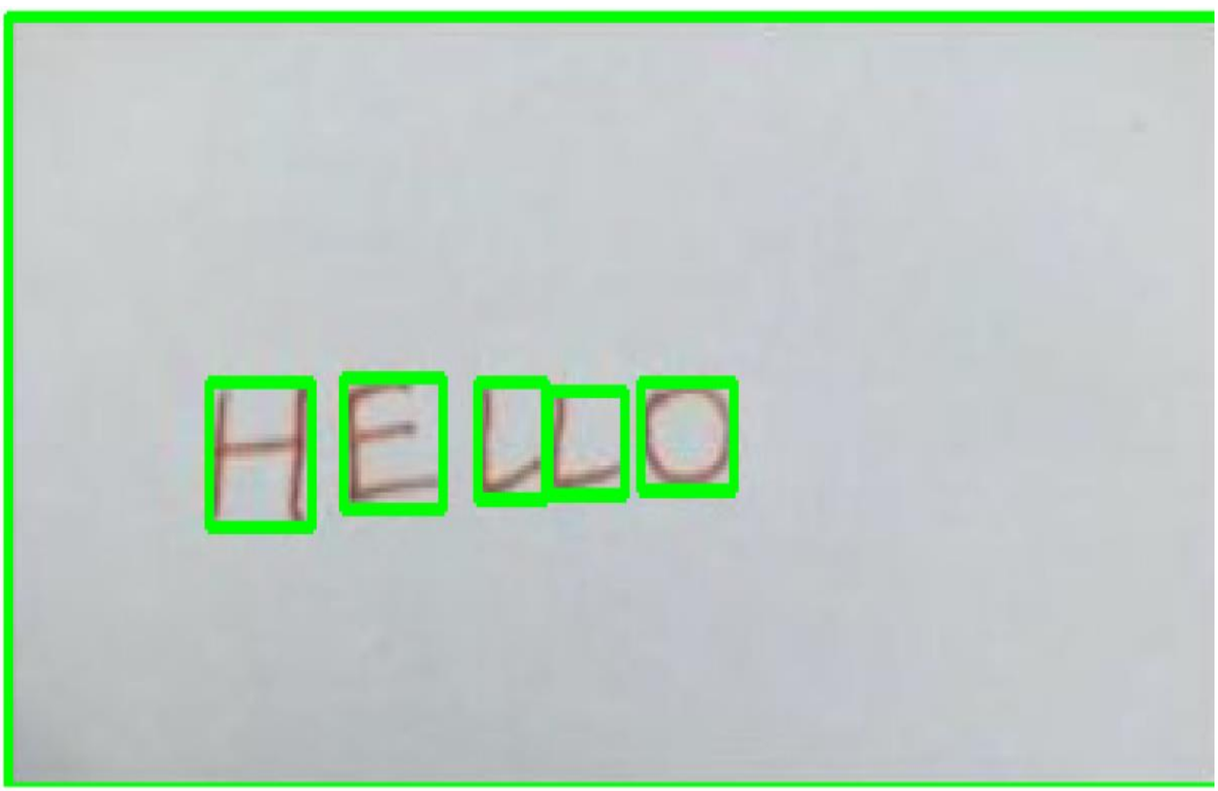
```python
# Case Study 2: License Plate Readers in Traffic Management
import cv2
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

model = tf.keras.models.load_model('Alphabet_Recognition.h5')
alpha_num = list('0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ')

def preprocess_char(char_img):
    char = cv2.resize(char_img, (28, 28))
    char = char.astype("float32") / 255.0
    char = char.reshape(1, 28, 28, 1)
    return char

def license_plate_reader(filepath):
    image = cv2.imread(filepath)

    if image is None:
        print("Image not found!")
        return

    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    blur = cv2.bilateralFilter(gray, 11, 17, 17)
    edged = cv2.Canny(blur, 30, 200)

    contours, _ = cv2.findContours(
        edged, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE
    )

    contours = sorted(contours, key=cv2.contourArea, reverse=True)[:10]

    plate = None

    for c in contours:
        peri = cv2.arcLength(c, True)
        approx = cv2.approxPolyDP(c, 0.02 * peri, True)

        if len(approx) == 4:
            x, y, w, h = cv2.boundingRect(c)
            plate = image[y:y+h, x:x+w]
            break
```

```python
    if plate is None:
        print("License plate not detected!")
        return

    gray_plate = cv2.cvtColor(plate, cv2.COLOR_BGR2GRAY)
    _, thresh = cv2.threshold(
        gray_plate, 0, 255,
        cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU
    )

    contours, _ = cv2.findContours(
        thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE
    )

    contours = sorted(contours, key=lambda c: cv2.boundingRect(c)[0])

    plate_text = []

    for c in contours:
        x, y, w, h = cv2.boundingRect(c)

        if h > 15 and w > 5:
            char_img = thresh[y:y+h, x:x+w]

            processed = preprocess_char(char_img)

            prediction = model.predict(processed, verbose=0)
            pred = alpha_num[np.argmax(prediction)]

            plate_text.append(pred)

    result = ''.join(plate_text)

    plt.figure(figsize=(8,3))
    plt.imshow(cv2.cvtColor(plate, cv2.COLOR_BGR2RGB))
    plt.axis('off')
    plt.show()

    print("License Plate:", result)


license_plate_reader("D:/EMNIST/Plate.jpg")
```

Case Study 3: Multilingual Handwritten Alphabet Recognition for Postal Automation
Scenario: A national postal service needs to automate mail sorting, including handwritten addresses written in multiple languages with different alphabets (Latin, Cyrillic, Arabic). The system must recognize handwritten characters accurately regardless of script or style variations.
Objective: Create a scalable neural network model capable of recognizing handwritten alphabetical characters from multiple languages, enabling automated and error-free sorting of mail across regions.

```python
# Case Study 3: Multilingual Handwritten Alphabet Recognition for Postal Automation
import cv2
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

model = tf.keras.models.load_model('Alphabet_Recognition.h5')

alpha = list('ABCDEFGHIJKLMNOPQRSTUVWXYZ')

def postal_address_recognition(filepath):

    image = cv2.imread(filepath)

    gray = cv2.cvtColor(
        image,
        cv2.COLOR_BGR2GRAY
    )

    thresh = cv2.adaptiveThreshold(
        gray,
        255,
        cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV,
        41,
        15
    )

    contours, _ = cv2.findContours(
        thresh,
        cv2.RETR_EXTERNAL,
        cv2.CHAIN_APPROX_SIMPLE
    )

    contours = sorted(
        contours,
        key=lambda c: cv2.boundingRect(c)[0]
    )

    address_text = []

    for c in contours:
```

```
            x, y, w, h = cv2.boundingRect(c)

            if w > 5 and h > 10:

                char = thresh[y:y+h, x:x+w]

                char = cv2.resize(char, (18,18))

                char = np.pad(
                    char,
                    ((5,5),(5,5)),
                    mode='constant'
                )

                prediction = model.predict(
                    char.reshape(1,28,28,1)/255.0,
                    verbose=0
                )

                pred = alpha[np.argmax(prediction)]

                address_text.append(pred)

                cv2.rectangle(
                    image,
                    (x,y),
                    (x+w,y+h),
                    (0,255,0),
                    2
                )

        plt.figure(figsize=(15,8))
        plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
        plt.axis('off')
        plt.show()

        print("Detected Address Text:")
        print("".join(address_text))

postal_address_recognition("D:/EMNIST/1.jpg")
```



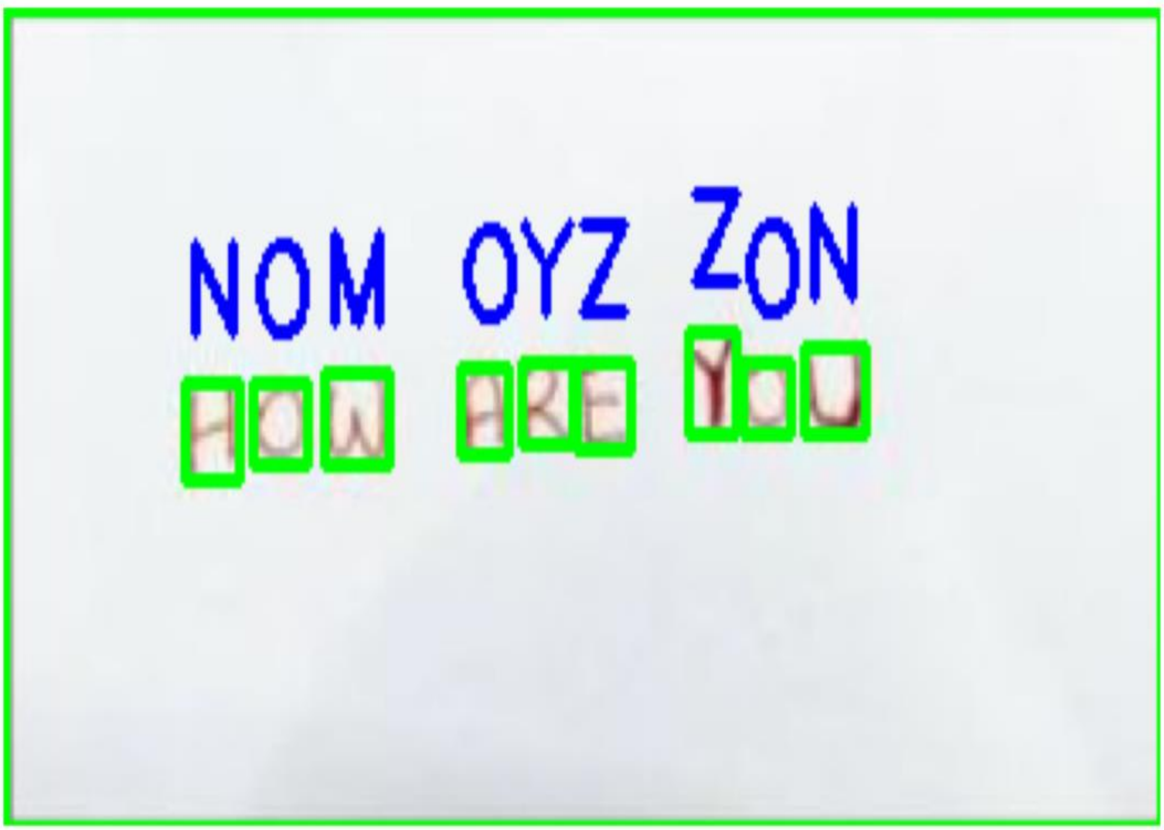
```
Detected Address Text:
FHONOO
```

Case Study 4: Real-Time Handwritten Input Recognition for Smart Devices
Scenario: A tech company is developing a note-taking application for tablets that converts handwritten input into editable digital text instantly. The app must handle diverse handwriting styles, quick strokes, and partial characters as the user writes naturally.
Objective: Implement a neural network-based handwritten alphabet recognition system optimized for real-time inference on mobile devices, balancing recognition accuracy with low latency.

```python
# Case Study 4: Real-Time Handwritten Input Recognition for Smart Devices
import cv2
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

model = tf.keras.models.load_model('Alphabet_Recognition.h5')

alpha = list('ABCDEFGHIJKLMNOPQRSTUVWXYZ')

def smart_device_handwriting(filepath):

    image = cv2.imread(filepath)

    gray = cv2.cvtColor(
        image,
        cv2.COLOR_BGR2GRAY
    )

    blur = cv2.GaussianBlur(
        gray,
        (3,3),
        0
    )

    thresh = cv2.adaptiveThreshold(
        blur,
        255,
        cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV,
        41,
        15
    )

    contours, _ = cv2.findContours(
        thresh,
        cv2.RETR_EXTERNAL,
        cv2.CHAIN_APPROX_SIMPLE
    )

    contours = sorted(
        contours,
        key=lambda c: cv2.boundingRect(c)[0]
```

```python
    )

    recognized_text = []

    for c in contours:

        x, y, w, h = cv2.boundingRect(c)

        if w > 5 and h > 10:

            char = thresh[y:y+h, x:x+w]

            char = cv2.resize(
                char,
                (18,18)
            )

            char = np.pad(
                char,
                ((5,5),(5,5)),
                mode='constant'
            )

            prediction = model.predict(
                char.reshape(1,28,28,1)/255.0,
                verbose=0
            )

            pred = alpha[np.argmax(prediction)]

            recognized_text.append(pred)

            cv2.rectangle(
                image,
                (x,y),
                (x+w,y+h),
                (0,255,0),
                2
            )

            cv2.putText(
                image,
                pred,
```

```
                pred,
                (x,y-10),
                cv2.FONT_HERSHEY_SIMPLEX,
                0.8,
                (255,0,0),
                2
            )

    plt.figure(figsize=(12,6))

    plt.imshow(
        cv2.cvtColor(
            image,
            cv2.COLOR_BGR2RGB
        )
    )

    plt.axis('off')

    plt.show()

    print("Recognized Text:")
    print("".join(recognized_text))

smart_device_handwriting("D:/EMNIST/2.jpg")
```

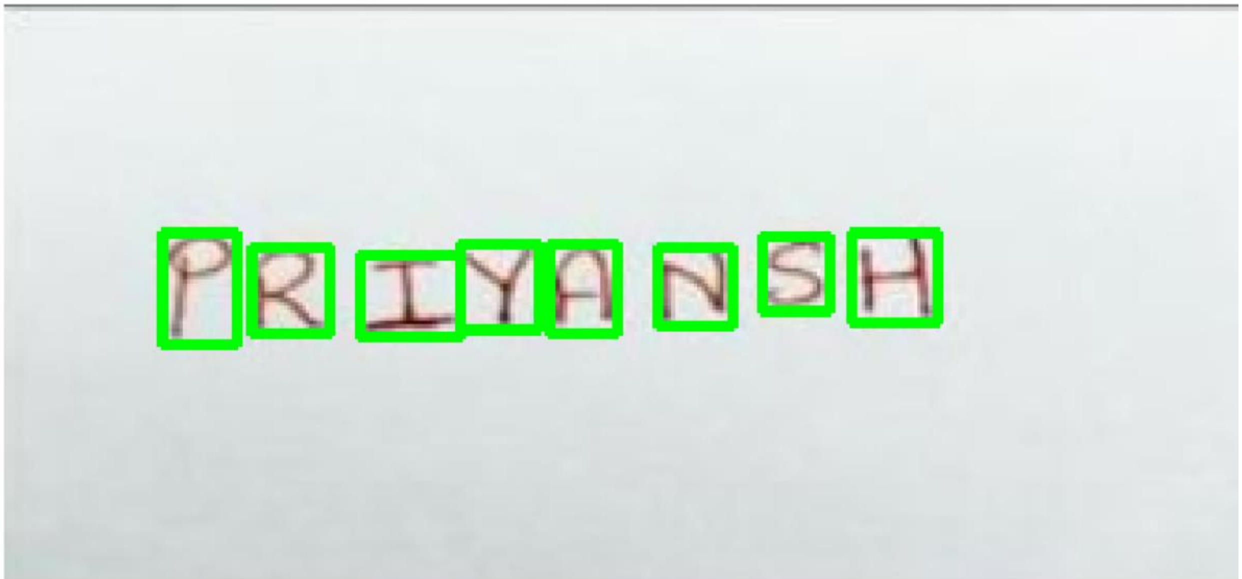



```
Recognized Text:
FNOMOYZZON
```

Case Study 5: Historical Manuscript Digitization
Scenario: A digital humanities research center aims to digitize and transcribe centuries-old handwritten manuscripts that contain faded, ornate, and irregularly styled alphabetic characters. Traditional OCR fails due to the antique script and degradation.
Objective: Develop a robust, adaptable handwritten alphabet recognition system using deep learning to assist in the automatic transcription of historical manuscripts, facilitating preservation and research.

```
# Case Study 5: Historical Manuscript Digitization
import cv2
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

model = tf.keras.models.load_model('Alphabet_Recognition.h5')

alpha = list('ABCDEFGHIJKLMNOPQRSTUVWXYZ')

def manuscript_transcription(filepath):

    image = cv2.imread(filepath)

    gray = cv2.cvtColor(
        image,
        cv2.COLOR_BGR2GRAY
    )

    enhanced = cv2.equalizeHist(gray)

    blur = cv2.GaussianBlur(
        enhanced,
        (5,5),
        0
    )

    thresh = cv2.adaptiveThreshold(
        blur,
        255,
        cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV,
        41,
        15
    )

    contours, _ = cv2.findContours(
        thresh,
        cv2.RETR_EXTERNAL,
        cv2.CHAIN_APPROX_SIMPLE
    )

    contours = sorted(
        contours,
        key=lambda c: cv2.boundingRect(c)[0]
    )

    manuscript_text = []

    for c in contours:

        x, y, w, h = cv2.boundingRect(c)

        if w > 5 and h > 10:

            char = thresh[y:y+h, x:x+w]

            char = cv2.resize(char, (18,18))

            char = np.pad(
                char,
                ((5,5),(5,5)),
                mode='constant'
            )

            prediction = model.predict(
                char.reshape(1,28,28,1)/255.0,
                verbose=0
            )

            pred = alpha[np.argmax(prediction)]

            manuscript_text.append(pred)

            cv2.rectangle(
                image,
                (x,y),
                (x+w,y+h),
                (0,255,0),
                2
            )

    plt.figure(figsize=(15,8))
    plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
    plt.axis('off')
    plt.show()
```

```
    print("Transcribed Manuscript:")
    print("".join(manuscript_text))

manuscript_transcription("D:/EMNIST/3.jpg")
```




```
Transcribed Manuscript:
DXIHBNSM
```

https://github.com/Mateen-Bashir